Hepatic Gluconeogenic and Ketogenic Interrelationships in the Lactating Cow[1]

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ABSTRACT

Interrelationships between propionate, palmitate, and butyrate metabolism were investigated in vitro with [1-carbon-14] carboxyl substrates. Production of labeled glucose, ketone bodies, and carbon dioxide was used to estimate rates of bovine hepatic gluconeogenesis and ketogenesis. Incubations were with liver slices from eight lactating Holstein cows fed either a control or high concentrate-low fiber diet. Liver samples were acquired by trochar biopsy at 30, 60, 90, and 180 days postpartum. Ketone production from both palmitate and butyrate was highest in liver slices obtained at 30 days. Glucose production from labeled propionate was also highest in early lactation. The higher rates of gluconeogenesis and ketogenesis in early lactation were associated with higher hepatic carnitine palmitoyltransferase (EC 2.3.1.21) activity. Feeding the high concentrate enhanced gluconeogenesis from propionate and decreased ketogenesis from palmitate. Propionate addition (10 m*M*) to incubation media also decreased the total amount of palmitate oxidized ([carbon-14] dioxide plus [carbon-14] ketones). Diet had no effect on hepatic butyrate metabolism. Results indicated that ketogenesis is regulated via rate of long chain fatty acid transport into the mitochondria. Stage of lactation has a greater influence on long and short chain fatty acid metabolism than does diet composition.

INTRODUCTION

Ruminant liver continually must synthesize glucose to meet requirements of maintenance and milk synthesis (37). Spontaneous ketosis can occur during peak lactation of dairy cows and just before parturition in sheep when demands for glucose and gluconeogenic substrates are greatest (11). The ketonemia characteristic of ketosis can be related to carbohydrate shortage, increased fatty acid mobilization from adipose, and increased uptake of long chain fatty acids by the liver. Similar conditions occur during periods of negative energy balance in ruminants (8) and nonruminants (15).

Control of ketogenesis in ruminant liver has been the subject of considerable debate. Excessive hepatic ketogenesis may result from lack of oxaloacetate necessary to condense with acetyl-CoA (20), indicating that tricarboxylic acid (TCA) cycle activity may be dependent on the supply and distribution of gluconeogenic precursors (3). Because ketotic cows and sheep are hypoglycemic (11), spontaneous ketosis is thought to be associated with conditions in which fatty acid concentration in blood increases without simultaneous increased availability of gluconeogenic precursors.

Hepatic ketogenesis in nonruminants is thought to be regulated by rate of transport of long chain fatty acids into the mitochondria (23, 24). Malonyl-CoA inhibition of long chain fatty acid oxidation is documented for nonruminants (22, 25). However, ruminant liver has limited capacity for fatty acid synthesis (5), and regulation of bovine ketogenesis via malonyl-CoA is questionable. Our objective was to investigate the relationship between hepatic gluconeogenesis and ketogenesis relative to rate of fatty acid transport in dairy cows.

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MATERIAL AND METHODS

Eight lactating Holstein cows were fed a control diet from 15 days to 60 days postpartum. After 60 days, cows were assigned randomly to two groups of four cows each. One group received a high concentrate-low fiber (HCLF) diet and the other continued to be fed the control diet until 180 days postpartum. Diet composition is in Table 1. Cows were housed in a free stall area during the experimental period and daily feed intake was recorded by a Pin-pointer feeder (UIS Corporation, Cookeville, TN). Liver samples were obtained by trochar biopsy at 30 days (n=4) and 60, 90, and 180 days (n=8) postpartum. Blood was sampled by jugular veinpuncture and ruminal fluid by stomach tube 3 days prior to each biopsy. Samples were obtained 2 h after feeding. Body weights were recorded weekly and milk weights daily. Milk samples (composites of two milkings) obtained 3 days before biopsy were analyzed for fat content (Multispec Infrared Analyzer, Berwind Instruments Ltd., York, England).

Plasma and Ruminal Fluid Analyses

Blood samples were analyzed for acetoacetate (AcAc) and D-(−)-β-hydroxybutyrate (BHB) as described by Williamson et al. (34). Glucose in blood plasma was assayed with glucose oxidase and peroxidase (Sigma Technical Bulletin No. 510, 1973). Insulin was measured by standard radioimmunoassay (2, 36). Ruminal fluid was analyzed for acetic, propionic, and butyric acids with 30 m*M* isocaproic acid as an internal standard by a Bendix gas chromatograph (Model 2600) as described by Supelco (Technical Bulletin No. 749B).

In Vitro Procedures

Rate of propionate metabolism to glucose and CO_2 and rate of palmitate and butyrate metabolism to ketones (AcAc and BHB) and CO_2 were determined with liver slices. Liver biopsy cores were cut on a Stadie-Riggs microtome to give slices weighing 75 to 125 mg. Slices were kept in cold calcium-free Krebs-Ringer bicarbonate buffer (pH 7.4) until blotted, weighed, and placed in 25-ml side arm flasks (Kontes, Vineland, NJ). Incubation media contained calcium-free Krebs-Ringer bicarbonate buffer, 5% (wt/vol) fatty acid-free bovine serum albumin and substrate in a total volume of 3 ml. Substrates were sodium salts of propionic acid (10 m*M*), palmitic acid (.4 m*M*), and butyric acid (5 m*M*). Radioactive tracers (.5 μCi/flask) were sodium salts of either [1-^{14}C] palmitic acid, [1-^{14}C] propionic acid, or [1-^{14}C] butyric acid (New England Nuclear, Boston, MA). Substrate and tracer combinations were: 1) palmitate, palmitate plus propionate and palmitate plus butyrate with [1-^{14}C] palmitate; 2) propionate and propionate plus palmitate with [1-^{14}C] propionate; and 3) butyrate and butyrate plus palmitate with [1-^{14}C] butyrate.

Immediately after addition of liver slice, each flask was gassed with oxygen:carbon dioxide (95:5) and capped with a rubber stopper from which a center well was suspended. The well contained a folded filter paper (2 × .2 cm) that was saturated with .2 ml of phenethylamine injected prior to termination. Incubations were terminated by acidification of the medium with 5% (vol/vol) perchloric acid injection (.5 ml). Incubations were in duplicate

TABLE 1. Percent compositions[1] of control and high concentrate-low fiber (HCLF) diets.

Ingredient	Control	HCLF
Corn silage	62.6	...
Ground shelled corn	...	46.6
Ground orchardgrass hay	7.9	23.3
Wet molasses	...	4.5
Protein supplement[2]	29.5	25.6

[1] Dry matter basis. Control and HCLF contained 23.0% and 15.5% acid detergent fiber, respectively.

[2] Protein supplements contained soybean meal, molasses, distillers dried grains, wheat, ground shelled corn, trace mineralized salt, vitamins A and D, and minerals in combinations calculated to supply 16.5% crude protein, .83% calcium, .51% phosphorus, 4800 IU vitamin A/kg, and 400 IU vitamin D/kg in both total mixed diets.

for 1 h at 37°C oscillating in a reciprocating water bath at 90 strokes/min. After an incubation was completed, labeled CO_2 was determined by the method described by Leng and Annison (16). Incubation medium was transferred to a test tube, neutralized with .1 ml of 20% (wt/vol) KOH, and centrifuged at 2600 × *g* for 15 min. Supernatant (2.0 ml) with .5 ml phosphate buffer (5 *M*; pH 6.8) added was frozen at −70°C.

Total radioactivity incorporated into glucose from [1-^{14}C] propionate was determined by eluting the medium on an anion exchange column (8 × 35 mm) of Dowex 1 × 8 (200 to 400 mesh) formate form. After application of 1 ml of medium to the column, glucose was eluted with 7.5 ml deionized water. Eluates were lyophylized and dissolved in 1 ml deionized water to determine radioactivity. For incubations in which [1-^{14}C] palmitic acid or [1-^{14}C] butyric acid were used, total [^{14}C] ketones were determined according to the procedure of McGarry et al. (23) employing aniline citrate and D-(−)β-hydroxybutyrate dehydrogenase (Sigma Chemical Co., St Louis, MO).

Carnitine Palmitoyltransferase

A portion of liver was frozen in liquid nitrogen immediately after each biopsy and stored at −70°C until analyzed for carnitine palmitoyltransferase and protein. Carnitine palmitoyltransferase (EC 2.3.1.21) activity was determined on isolated mitochondria by the isotope-forward method described by Norum (27). Liver protein concentrations were determined by method of Lowry (18).

Statistical Analysis

Results from liver slices of cows fed the control diet were compared at 30, 60, 90, and 180 days by orthogonal contrasts of the respective least square means for determination of stage of lactation and substrate effects. Because linear contrasts across stages of lactation within each substrate combination and interactions between stage and substrate were not significant ($P<.05$), least square means containing all stages of lactation were used to test substrate effects, and least square means for all substrate combinations were used to test effects of stage of lactation. Incubation, blood, milk, and ruminal fluid data from all cows at 60, 90, and 180 days postpartum were analyzed by covariate analysis to determine diet effects with the 60-day observations the covariate. Significant differences ($P<.05$) between least square means were determined by the Tukey test.

RESULTS AND DISCUSSION

Feeding the HCLF diet had no significant effect on milk production or body weight (Table 2). However, milk fat percent was lower ($P<.05$) for cows fed HCLF. Depressed milk fat percent has occurred most often in cows fed excess concentrate in late lactation (13, 32). Central to theories of depressed milk fat percent is increased ruminal propionate production. Feeding rations containing more than 60% concentrate increased ruminal propionate (31, 32). No differences in ruminal acetate, propionate, or butyrate concentrations were observed in this study. Because ruminal fluid was collected 2 h after morning feeding, maximum volatile fatty acid production may not have occurred (7), and the observed concentrations may not be appropriate indicators of actual volatile fatty acid production or availability.

Concentrations of insulin, glucose, and ketones (AcAc plus BHB) were not significantly different between diets (Table 2). However, insulin did tend to be higher and ketone concentrations lower for cows fed the HCLF diet. In both groups of cows insulin and glucose tended to increase as lactation progressed and milk production decreased. Results by other investigators were similar (2, 12, 17).

Stage of Lactation Effects on Hepatic Fatty Acid Metabolism

Effects of stage of lactation on rate of glucose and CO_2 production from [1-^{14}C] propionate and [1-^{14}C] palmitate and [1-^{14}C] butyrate metabolism to ketone bodies and CO_2 are in Table 3. The in vitro hepatic output of glucose from [1-^{14}C] propionate was greater ($P<.05$) at 30 and 60 days than at 90 and 180 days postpartum. Additionally, incorporation of labeled carbon into CO_2 decreased curvilinearly with days in lactation. Bauman and Currie (6) reviewed the importance of an increased gluconeogenic rate associated with initiation of lactation. Mathias and Elliott (21) using liver homogenates were unable to show an

TABLE 2. Effects of diet and stage of lactation on milk yield, milk fat, body weight, feed intake, rumen volatile fatty acid (VFA) concentrations, and plasma glucose, ketone, and insulin concentrations.

		Stage of lactation			Stage of lactation		
	Diet	30 days	60 days	Diet	90 days	180 days	SE
Cows, n[1]	Control	4	4	Control	4	4	
			4	HCLF	4	4	
Milk, kg/day	Control	28.3	26.0	Control	23.5	20.0	1.5
			29.2	HCLF	29.7	23.2	2.5
Milk fat, %[2]	Control	3.58	3.23	Control	3.47	3.76	.2
			3.24	HCLF	3.03	3.22	.16
Body weight, kg	Control	581	564	Control	567	628	47
			543	HCLF	556	593	31
Dry matter intake kg/day[3]	Control	16.1	16.4	Control	17.9	18.2	.9
			18.8	HCLF	23.3	19.2	1.6
Rumen VFA							
Acetate, μmol/ml	Control	59.9	59.4	Control	59.8	80.9	10.3
			73.3	HCLF	61.6	66.2	12.4
Propionate, μmol/ml	Control	26.4	26.5	Control	28.3	31.6	5.5
			31.5	HCLF	29.4	32.3	4.8
Butyrate, μmol/ml	Control	15.2	13.7	Control	13.7	14.9	4.0
			17.5	HCLF	12.9	14.1	2.8
Plasma							
Glucose, mg/dl	Control	65.0	65.1	Control	66.7	68.9	1.5
			64.6	HCLF	67.8	69.6	2.2
Ketones, μmol/ml[4]	Control	.34	.36	Control	.40	.30	.04
			.35	HCLF	.30	.27	.04
Insulin, ng/ml	Control	.59	.61	Control	.73	.80	.12
			.71	HCLF	.83	.94	.16

[1] Two cows from each group were biopsied at 30 days. All cows were biopsied at 60 days. High concentrate-low fiber (HCLF) feeding began at 61 days. Data are least square means.

[2] Significant difference ($P<.05$) between diets at 90 and 180 days.

[3] Significant difference ($P<.05$) between diets at 90 days.

[4] Acetoacetate and β-hydroxybutyrate.

increase of propionate metabolism in early lactation. Gluconeogenic capacity determined by hepatic glucose output, however, was greater for lactating than for nonlactating cows (4).

Several workers have hypothesized that the rate of hepatic ketogenesis is determined by the supply of oxaloacetate relative to its production from propionate and utilization for glucose synthesis (3, 6, 20). Ketogenesis from both [1-^{14}C] palmitate and [1-^{14}C] butyrate was maximum in 30-day liver slices, similar to labeled glucose production from [1-^{14}C] propionate (Table 3). These results suggest that rates of hepatic gluconeogenesis and ketogenesis are closely related under normal conditions. The decreased ability to oxidize palmitate to ketones in later lactation was probably the result of decreased carnitine palmitolytransferase activity as lactation progressed (Figure 1). McGarry and coworkers (24, 25) suggested that states of negative energy balance in nonruminants would increase the ability of hepatic tissue to oxidize long chain fatty acids. They further suggested a low ratio of insulin to glucagon would relieve the allosteric inhibition of carnitine palmitoyltransferase by malonyl-CoA. One feature of early lactation in cows is that blood concentration of insulin and insulin

TABLE 3. Effect of stage of lactation on in vitro hepatic metabolism.[1]

Substrate and Products	Stage of Lactation (days) 30	60	90	180	SE
[1-^{14}C] Propionate[2]					
CO_2	117^a	88^b	75^{bc}	63^c	8
Glucose	17^a	12^b	5^c	5^c	2
[1-^{14}C] Palmitate[3]					
CO_2	.20^a	.20^a	.24^b	.25^b	.02
Ketones	.70^a	.63^{ac}	.45^b	.54^c	.04
CO_2/ketones	.29^a	.31^a	.55^b	.59^b	.04
[1-^{14}C] Butyrate[4]					
CO_2	56^a	40^b	38^b	36^b	6
Ketones	77^a	58^b	51^b	55^b	6
CO_2/ketones	.78	.69	.75	.62	.09

[1] Liver slices were incubated in Krebs-Ringer bicarbonate buffer (pH 7.4) with combinations of Na-palmitate (.4 m*M*), Na-propionate (10 m*M*) and Na-butyrate (5 m*M*). Results are [^{14}C] substrates metabolized to [^{14}C] product (nmol·mg protein^{-1} · h^{-1}). Data are least square means from cows fed the control diet.

[2] Combined data from propionate and propionate plus palmitate.

[3] Combined data from palmitate, palmitate plus propionate and palmitate plus butyrate.

[4] Combined data from butyrate and butyrate plus palmitate.

[a,b,c] Means across stages of lactation with different superscripts differ ($P<.05$).

secretory responsiveness are low. Aiello et al. (2) found a low ratio of insulin to glucagon in early lactation associated with negative energy balance. Because activities of enzymes responsible for fatty acid synthesis in ruminant liver (5) are low, malonyl CoA may not be produced in sufficient quantities to act as a supressor of ketogenesis, and the ratio of insulin to glucagon may serve as a primary regulator.

Effects of Diet on Hepatic Fatty Acid Metabolism

Labeled glucose production from [1-^{14}C] propionate was greater ($P<.05$) for HCLF compared to controls (Table 4). Baird et al. (3) found increased uptake of propionate by hepatic tissue when propionate was infused into the portal vein. Ricks and Cook (28) provided evidence that the Michaelis-Menten constant (K_m) of bovine liver propionyl-CoA synthetase is regulated by concentration of propionate in portal blood.

Effects of HCLF feeding on palmitate metabolism also are in Table 4. Relative to cows fed the control ration, incorporation of radioactive carbon from [1-^{14}C] palmitate into CO_2 was lower ($P<.05$), and label incorporation into ketones tended to be lower in liver slices from the HCLF cows. In contrast to the positive relationship between gluconeogenesis and ketogenesis with stage of lactation (Table 3), feeding the HCLF ration apparently caused the relationship to be reversed. The total amount of palmitate oxidized (CO_2 plus ketones) was decreased ($P<.05$) in liver from cows fed the HCLF diet while glucose production from propionate was increased.

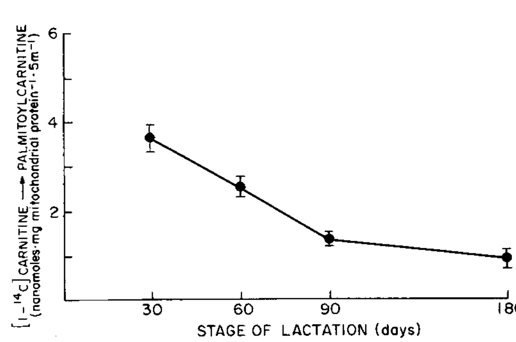

Figure 1. Mitochondrial carnitine palmitoyltransferase activity in liver of lactating cows biopsied at several stages of lactation.

TABLE 4. Effects of a high concentrate-low fiber (HCLF) diet on in vitro hepatic metabolism.[1]

Substrate	Diet	Products					
		CO_2		Glucose			
			SE		SE		
[1-^{14}C] Propionate[2]	Control	67	6	4.1*	8		
	HCLF	69		6.5			
		CO_2		Ketones		(CO_2 plus ketones)	
			SE		SE		SE
[1-^{14}C] Palmitate[2]	Control	.29*	.01	.47	.03	.76*	.04
	HCLF	.18		.39		.58	
		CO_2		Ketones		(CO_2 plus ketones)	
			SE		SE		SE
[1-^{14}C] Butyrate[2]	Control	35	4	55	5	90	6
	HCLF	39		48		86	

[1] Results are expressed as [^{14}C] substrate metabolized to [^{14}C] product (nmol·mg protein^{-1} · h^{-1}). Data are least square means.

[2] Combined data from 90 and 180 days.

*Significant difference ($P<.05$) between diets.

Diets high in digestible carbohydrates elevated insulin concentration in plasma (12). Decreased palmitate oxidation may have resulted from liver of cows fed HCLF being predisposed to higher concentrations of insulin prior to biopsy. Insulin concentration tended to be higher in these animals (Table 2); however, a direct regulatory role of insulin in hepatic ketogenesis has not been determined for ruminants.

Several investigators have concluded that the extent of ketone body production might be related to quantity of acetyl-CoA available for condensation to acetoacetyl-CoA (29, 35). The ability of glucose administration to depress hepatic ketogenesis in goats was attributed to decreasing the acetyl-CoA pool by decreasing long chain fatty acid oxidation (26). Control of partitioning of long chain fatty acids between oxidation and reesterification via changes of carnitine palmitoyltransferase activity is documented for nonruminants (22, 24). As support for similar control in ruminants, Snoswell and Koundakjian (30) reported increased carnitine palmitoyltransferase activity in the liver of starved and alloxan diabetic sheep. Our data also suggest regulation of ketogenesis prior to beta-oxidation, possibly via carnitine palmitoyltransferase. Sufficient tissue was not available to determine carnitine palmitoyltransferase activity in all HCLF cows at 180 days; however, 90-day activity was lower ($P<.05$) in liver of HCLF cows.

In contrast to findings with [1-^{14}C] palmitate, diet had no effect on [1-^{14}C] butyrate metabolism to carbon dioxide or ketone bodies (Table 3). Butyryl-CoA synthetase activates butyrate to the acyl-CoA form within the mitochondria, independent of a mitochondrial transport system (1).

Substrate Effects

Effects of butyrate and propionate on [1-^{14}C] palmitate metabolism are in Figure 2. Both butyrate and propionate were expected to contribute additional carbon to the acetyl-CoA pool and, as such, decrease specific radioactivity of the pool causing decreased ^{14}C in CO_2 and ketones. In vitro studies (14) with [1-^{14}C] and [2-^{14}C] propionate indicated that 45 to 65% of propionate carbon is converted to glucose; therefore, propionate was added at twice the concentration of butyrate in an attempt to equalize the effect on the acetyl-CoA pool. Addition of butyrate caused a non-

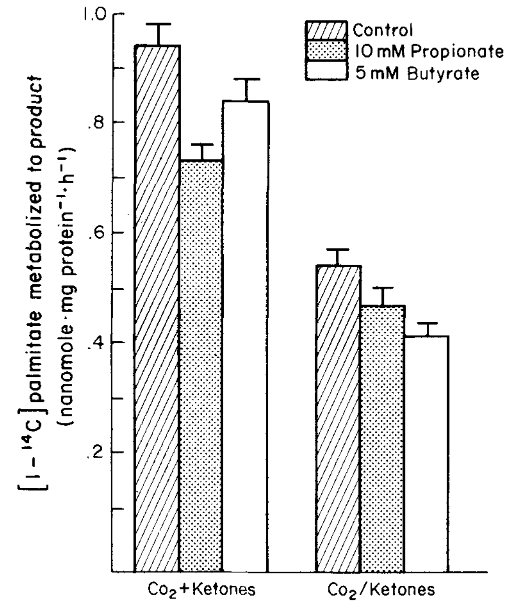

Figure 2. Effects of propionate and butyrate additions to incubation media on in vitro palmitate oxidation by lactating cows liver slices.

significant decrease of the total amount of labeled carbon dioxide and ketones produced. The ratio of labeled CO_2 to ketones, however, was lowered ($P<.05$) by added butyrate, suggesting a ketogenic effect. Ketosis often is considered to be caused by conditions that increase the acetyl-CoA pool and reduce the amount of acetyl-CoA oxidized by the TCA cycle (3, 5, 9, 35).

Addition of 10 m*M* propionate decreased ($P<.05$) total labeled ketone and carbon dioxide production from [1-^{14}C] palmitate compared to control. Although a portion of the decrease may have been from decreased acetyl-CoA pool specific radioactivity, other factors may have contributed to the decreased palmitate utilization. Bush and Milligan (10) suggested that propionyl-CoA inhibits 3-hydroxy-3-methylglutaryl-CoA (HMG-CoA) synthase (EC 4.1.3.5) and thus decreases ketogenesis. The effect of inhibiting HMG-CoA synthase would be to increase the ratio of CO_2 to ketones without decreasing the total amount of palmitate oxidized. Because the ratio of CO_2 to ketones was not changed significantly and palmitate oxidation was decreased, HMG-CoA synthase activity apparently was not influenced by addition of propionate to the media.

Competition between fatty acids for the available CoA may be an explanation for our results. The importance of the availability of CoA for activation and oxidation of fatty acids in nonruminants has been demonstrated by Weidiman and Krebs (33). The low K_m reported by Ricks and Cook (28) for propionyl-CoA synthetase in ruminant liver indicated a high affinity for propionate and CoA. They also suggested that when propionate concentration exceeds the K_m, other acyl-CoA synthetases activate propionate to ensure uptake for glucose synthesis. This process may have limited CoA availability for long chain fatty acid transport into mitochondria.

The antiketogenic effect of propionate has been attributed to decreasing the supply of free fatty acids from adipose to the liver (32). Propionate and butyrate are stimulators of insulin secretion in ruminants (19). Several researchers emphasized the importance of insulin as an antiketogenic mechanism via increased triglyceride synthesis and decreased fatty acid mobilization. However, hormonal regulation cannot account for effects of propionate and butyrate reported herein. Incubation media contained a constant amount of palmitate (.4 m*M*) and contained no added hormone.

Summary

The position of carnitine palmitoyltransferase in fatty acid metabolism makes it a primary site for regulation of hepatic ketogenesis, and nonruminant studies have concluded that hepatic output of ketone bodies is correlated with carnitine palmitoyltransferase activity. This study provides evidence that carnitine palmitoyltransferase is affected by stage of lactation. An HCLF diet decreased carnitine palmitoyltransferase activity and palmitate oxidation but not butyrate oxidation. In vitro addition of propionate also decreased palmitate oxidation. Research directed towards kinetic properties and hormonal control of bovine hepatic carnitine palmitoyltransferase may be useful in elucidating factors controlling ruminant ketogenesis and related changes of gluconeogoenesis.

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